6.7 Polar Coordinates

Pre-Calculus 6.7 · Student sheet

Name _____ Period ____ Date _____

I can…

- plot points and give several polar representations of a point.
- convert points between polar and rectangular form.
- convert equations between polar and rectangular form.

Vocabulary · fill in the missing word

Pole — the origin of a polar _____ system

Polar axis — the ray from the pole along the _____ x-axis

Polar coordinates — (r, θ): r is the _____ distance from the pole and θ is the directed angle from the polar axis

Part 1 · Many names for one point

Plot each point on polar graph paper.

Point	(2, π/3)	(2, 7π/3)	(−2, 4π/3)	(2, −5π/3)
Where it lands				

Part 2 · Notes and examples (fill in each note, then work the example)

Polar coordinates (r, θ): r is the directed distance from the _____; θ is the angle from the polar axis.

Example 1

Plot $(2, \frac{\pi}{3})$, $(3, -\frac{\pi}{6})$, and $(-1, \frac{3\pi}{4})$.

Multiple representations: (r, θ) = (r, θ + 2nπ) = (_____, θ + (2n + 1)π)

Example 2

Give three other representations of $(3, -\frac{3\pi}{4})$ with $-2\pi < \theta < 2\pi$.

Conversions: x = r cos θ, y = r sin θ; $r^2 = x^2 + y^2$, tan θ = _____

Example 3

Convert $(\sqrt{3}, \frac{\pi}{6})$ to rectangular coordinates.

Example 4

Convert $(-1, 1)$ to polar coordinates.

Example 5

Convert.

a. $r = 2$ to rectangular

b. $\theta = \frac{\pi}{3}$ to rectangular

c. $r = \sec\theta$ to rectangular

d. $x^2 + y^2 = 9$ to polar

Part 3 · Partner practice

Partner 1

Convert $(4, \frac{2\pi}{3})$ to rectangular coordinates.

Partner 2

Convert $(0, -5)$ to polar coordinates.